Movie Memory Test: Effects of Boundary Type on Recognition, Signal Detection, and Metacognition

BRSM PROJECT POSTER PRESENTATION SP 2026

ABSTRACT

This study investigates how boundary type (Abrupt Cut vs. Natural Cut) and target frame type (Event-Model vs. Boundary-Break) affect recognition memory for movie clips. 170 participants viewed 40 clips and completed a 2AFC recognition task. We found that natural event segmentation preserves memory encoding ($d = 0.41$), while the predicted boundary advantage (BB $>$ EM) was not observed. Signal Detection Theory confirmed the NB advantage reflects genuine discriminability ($d' = 2.26$ vs. 2.01). Abrupt boundaries selectively impair metacognitive certainty for boundary-adjacent content while accuracy remains comparable.

METHODOLOGY

Design: 2 (Boundary Type: AB vs. NB; between) $\times$ 2 (Target Type: EM vs. BB; within) mixed design.

Analytical Pipeline:

- Descriptive statistics and assumption checks (Shapiro-Wilk, Levene's)
- 2$\times$2 Mixed ANOVA with η_p^2
- Non-parametric robustness checks (Mann-Whitney, Wilcoxon)
- Signal Detection Theory: d' and criterion c with log-linear correction
- Mixed-effects models with crossed random effects (subjects $\times$ items)
- Demographic moderation (ANCOVA)
- Bayesian analysis for near-significant effects

Missing Data: Demographics for 22 participants imputed using **median** (age) and **mode** (categoricals).

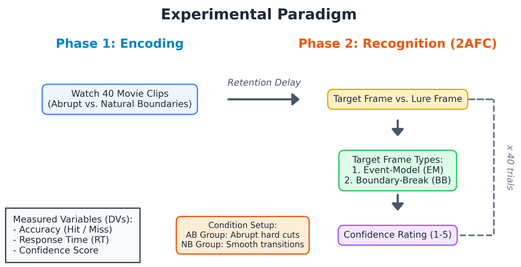

Figure 1: Experimental paradigm — encoding → 2AFC recognition → confidence.

RESULTS

H1 (NB $>$ AB Accuracy): Supported. Mixed ANOVA: $F(1, 168) = 7.25$, $p = .008$, $\eta_p^2 = .041$. NB ($M = .871$) outperformed AB ($M = .840$), Cohen's $d = 0.41$ (medium effect). Non-parametric Mann-Whitney confirmed: $U = 2744$, $p = .007$.

H2 (BB $>$ EM Boundary Advantage): Not Supported. $F(1, 168) = 11.44$, $p < .001$, $\eta_p^2 = .064$. **Direction reversed**: EM ($M = .870$) $>$ BB ($M = .843$). The boundary advantage (Radvansky & Zacks, 2017) does not hold when boundaries are artificially disrupted.

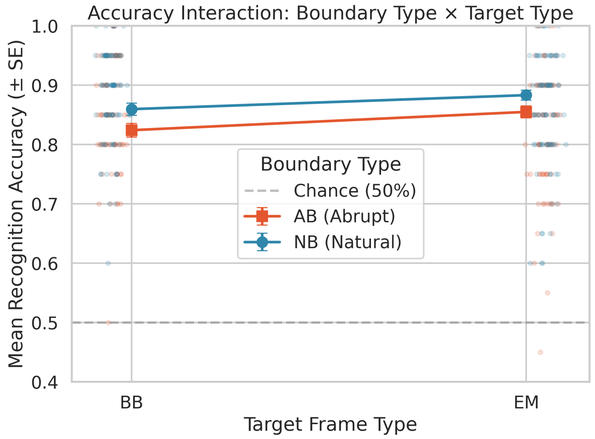

Figure 2: Accuracy interaction with individual data.

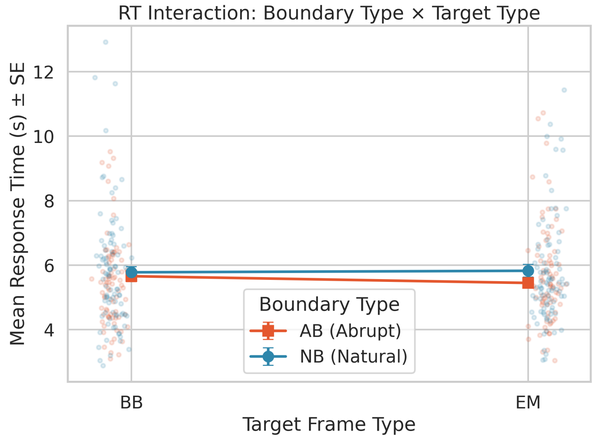

Figure 3: RT interaction with individual data.

H4 (RT Interaction): Partially Supported. Omnibus: $F(1, 168) = 3.36$, $p = .069$. Simple effects: AB group was significantly slower for BB than EM ($t = -2.56$, $p = .012$, $d = -0.15$); NB showed no difference ($p = .67$). Bayesian $\mathrm{BF}_{10} = 0.78$ (anecdotal evidence).

H5 (SDT d'): Supported. NB $d' = 2.26$ vs. AB $d' = 2.01$; $t = 2.55$, $p = .012$, $d = 0.39$. The NB advantage reflects genuine discriminability (criterion c is undefined in 2AFC).

Confidence Interaction

Significant interaction $F(1, 168) = 4.03$, $p = .046$.

- AB: EM $>$ BB ($p = .003$, $d = .19$)
- NB: EM $\approx$ BB ($p = .733$)
- BB: NB $>$ AB ($p = .024$, $d = .35$)

Abrupt cuts impair **metacognitive certainty** for boundary-adjacent content.

Mixed-Effects Models

Crossed random effects (subjects $\times$ items):

- Condition: $b = .036$, $z = 2.95$, $p = .003$
- Target Type: $b = .031$, $z = 2.51$, $p = .012$
- Interaction: $b = -.007$, $p = .669$

All effects robust to item-level variability.

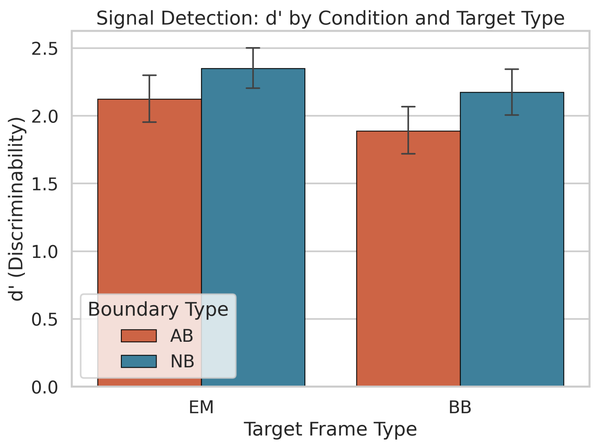

Figure 4: SDT d' (discriminability) by condition.

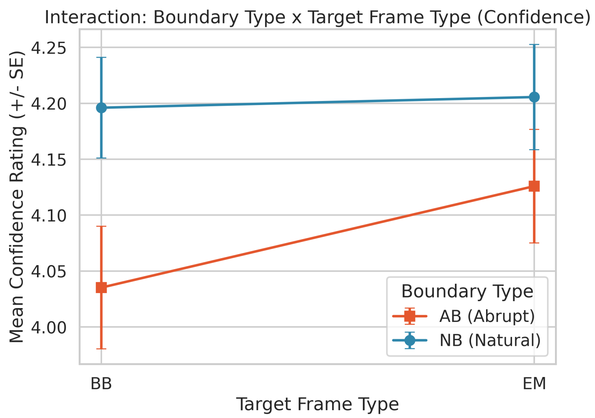

Figure 5: Confidence interaction plot.

CONCLUSION & FUTURE WORK

Key findings:

1. **Natural boundaries preserve memory**: NB group showed higher accuracy ($d = 0.41$) and discriminability ($d' = 0.39$), supporting Event Segmentation Theory.
2. **Boundary advantage fails** when boundaries are artificially disrupted (H2 rejected).
3. **Metacognitive dissociation**: AB group was less confident about BB targets while maintaining comparable accuracy — abrupt cuts impair subjective certainty.
4. **RT interaction**: AB participants selectively slow down for BB targets, consistent with disrupted event model updating.
5. **Demographics**: Age negatively correlated with accuracy ($\rho = -.203$, $p = .008$); effect survives ANCOVA.

Future directions:

- Yes/no recognition for richer SDT
- Model item-level predictors
- Replicate RT interaction
- Within-subjects boundary design

HYPOTHESIS SUMMARY

H#	Prediction	Result
H1	NB $>$ AB acc.	**Supported**
H2	BB $>$ EM	**Rejected**
H3	Acc. interact.	Not sig.
H4	RT interact.	**Partial**
H5	NB $d' >$ AB	**Supported**
H6	Demographics	**Partial**

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